

Zhenghong Li 李政宏

Physics Undergraduate, Shanghai Jiao Tong University

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Education

Shanghai Jiao Tong University

2024 - 2028 (expected)

Tsung-Dao Lee Physics Class; Physics (Strong Foundation Program), School of Physics and Astronomy

- All courses: average score 91.26/100 (rank 3/50); GPA 3.951 (rank 3/50).
- Core courses: average score 91.06/100 (rank 5/50); GPA 3.936 (rank 3/50).

Shenzhen Middle School

2021 - 2024

Shenzhen, Guangdong, China

Research Experience

Research advisors: Hong Ding (Academician) & Baiqing Lv (Assoc. Prof.)

Tsung-Dao Lee Institute, Shanghai Jiao Tong University

ARPES with In Situ Electric and Magnetic Fields

Designed and built a magnetic-field ARPES system and acquired spectra up to 50 mT; designed an integrated electric/magnetic-field sample holder and modeled field distributions and photoelectron collection in COMSOL to optimize collection.

Experiment-Guided, Self-Improving AI-DFT Agent

Build an AI-DFT agent workflow that cross-checks DFT calculations with ARPES, STM, and optical measurements; expand from simple systems to complex materials and refine calculation strategies and physical interpretation using experimental feedback.

FrontierPhysics: AI Physics Research Benchmark

Contribute to benchmark task design and iteration; review derivations, model assumptions, and numerical results in other tasks; evaluate AI physical reasoning, noisy-data inference, and experimental decisions. Related work submitted to ICLR.

Nickelate Superconductors: Zhiyuan Future Scholars Project

Project lead; funding: RMB 100,000. Explore substrate preparation and thin-film growth; characterize surface morphology, structure, and transport using AFM, RHEED, XRD, and low-temperature electrical transport measurements.

Honors and Awards

- Shanghai Municipal Scholarship
- SJTU Undergraduate Excellence Scholarship, Grade B
- Champion, SJTU RoboCup Small Size League Technical Challenge
- COPT-Sponsored Special Optimization Award, Celian Cup Mathematical Modeling Elite League
- Third Prize, HelloHPC Supercomputing Cluster Optimization Competition

Selected Projects and Activities

- **Academic and research exploration:** Participated in an AI4Science hackathon and presented an agent for predicting quantum material properties. Attended the 2026 CLS-CIBR-IDG Neuroscience and Cognitive Science Summer School at Peking University, TDLI's Tianwen Undergraduate Academic Forum, and SJTU's Jiao · Chuang Academic Forum.
- **Social service and student activities:** Served in the SJTU Youth League Organization Department and SEIEE Volunteer Team; organized and coordinated major events, including Shanghai Marathon volunteering and university Youth League induction ceremonies.

Skills and Languages

Skills: ARPES, COMSOL, AI For Science work in physics.

TOEFL: 105 (R: 28, L: 25, S: 25, W: 27; Mar 2025).

CET: CET-4: 651 (Dec 2024); CET-6: 594 (Jun 2025).